

Graph Search Algorithms

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1 Free Primitives

Treat any algorithm with linear running time as something we can essentially use for free.

2 Generic Graph Search

Input: An undirected/directed graph $G = (V, E)$, and a starting vertex $s \in V$

Output: Identify the vertices of V reachable from s in G .

Generic Graph Search
Input: graph $G = (V, E)$ and vertex $s \in V$ Output: a vertex is reachable from s if and only if it is marked as “explored”
mark s as explored and all others unexplored while $\exists (v, w) \in E$ with v explored and w unexplored do choose some edge (v, w) mark w as explored

3 Breadth First Search

Explores layer by layer, Layer 0 being s itself.

Breadth First Search (BFS)
mark s as explored, all other vertices as unexplored $Q \leftarrow$ a queue data structure, init with s while Q is not empty do remove the vertex from the front of Q , call it v foreach edge (v, w) in v 's adjacency list do if w is unexplored then mark w as explored add w to the end of queue

Runtime - $O(m + n)$

4 Connected Components

In an undirected graph $G = (V, E)$, a maximal subset $S \subseteq V$ of vertices such that there is a path from any vertex in S to any other vertex in S .

UCC algorithm is used which takes undirected graph $G = (V, E)$ in adjacency-list representation with $V = \{1, 2, 3, \dots, n\}$

Postcondition: $\forall u, v \in V, cc(u) = cc(v) \iff u, v$ are in the same connected component

Undirected Connected Components (UCC)	
<pre> mark all vertices as unexplored numCC $\leftarrow$ 0 for $i \leftarrow 1$ to n do if i <i>is unexplored</i> then numCC $\leftarrow$ numCC + 1 // new component // call BFS $Q \leftarrow$ a queue data structure, init with s while Q <i>is not empty</i> do remove the vertex from the front of Q, call it v foreach $edge(v, w)$ <i>in</i> v's adjacency list do if w <i>is unexplored</i> then mark w as explored add w to the end of queue </pre>	

Runtime - $O(m + n)$

5 Depth First Search

Takes one path and only backtracks when absolutely necessary.

Iterative Implementation of DFS	
<pre> mark all vertices as unexplored $S \leftarrow$ a stack data structure, init with s while S <i>is not empty</i> do pop the vertex v from the front of S if v <i>is unexplored</i> then mark v as explored foreach $edge(v, w)$ <i>in</i> v's adjacency list do push w to the front of S </pre>	

Recursive Implementation of DFS
<pre> Func DFS(G, v): mark s as explored and all other as unexplored foreach $edge(s, v)$ in s's adjacency list do if v is unexplored then DFS(G, v) </pre>

Runtime - $O(m + n)$

6 Topological Sort

Topological Ordering

Let $G = (V, E)$ be a directed graph. A topological ordering of G is an assignment $f(v)$ of every vertex $v \in V$ to a different number such that

$$\forall (v, w) \in E, f(v) < f(w)$$

- Topological orderings do not exist in directed cyclic graphs.
- Every DAG has a topological ordering.

Topological Sort

DAG $G = (V, E)$ in adjacency list representation is given a f value for every vertex constituting a topological ordering. It uses DFS-Topo which assigns every vertex reachable from a vertex s an f -value.

TopoSort
<pre> Func TopoSort(G): mark all vertices as unexplored $curLabel \leftarrow V$ foreach $v \in V$ do if v is unexplored then DFS-Topo(G, v) </pre>

DFS-Topo
<pre> Func DFS-Topo(G, s): mark s as explored foreach $edge(s, v)$ in s's outgoing adjacency list do if v is unexplored then DFS-Topo(G, v) $f(s) \leftarrow curLabel$ $curLabel \leftarrow curLabel - 1$ </pre>

$curLabel$ is a global variable, initialised to $|V|$. Runtime is $O(m + n)$.

7 Strongly Connected Components

A strongly connected component or SCC of a directed graph is a maximal subset $S \subseteq V$ of vertices such that there is a directed path from any vertex in S to any other vertex in S .

Every directed graph can be viewed as directed acyclic graph built up from its SCCs.

The SCC Meta-Graph is Directed Acyclic

Let $G = (V, E)$ be a directed graph. Define the corresponding meta-graph $H = (X, F)$ with one meta-vertex $x \in X$ per SCC of G and a meta-edge (x, y) in F whenever there is an edge in G from a vertex in the SCC corresponding to x to one in the SCC corresponding to y . Then H is a directed acyclic graph.

Therefore, it is basically a compressed graph where each SCC of the graph is compressed into a node.

7.1 Kosaraju

Corollary: For a directed graph G , let $f(v), g(v)$ denote position of $v \in V$ as computed by TopoSort algorithm for G and G^{rev} respectively. Also, let S_1, S_2 denote two SCCs of G with an edge (v, w) with $v \in S_1$ and $w \in S_2$.

$$\min_{x \in S_1} f(x) < \min_{y \in S_2} f(y).$$

$$\min_{x \in S_1} g(x) > \min_{y \in S_2} g(y).$$

We can see that the vertex x with $f(x) = 1$ will be the source vertex i.e for two connecting SCCs the one with x will be the source of the directed edge. Therefore, for G^{rev} , it corresponds to a vertex y such that $g(y) = 1$ and y resides in the 'sink' SCC. (Sink SCC is the SCC that connects to no other SCC). It is vital to identify these sink SCCs for Kosaraju.

Kosaraju
Input: directed graph $G = (V, E)$ in adjacency-list representation, with $V = \{1, 2, 3, \dots, n\}$ Output: for every $v, w \in V$, $scc(v) = scc(w) \iff v, w$ are in the same SCC of G $G^{rev} \leftarrow G$ with all edges reversed mark all vertices of G^{rev} as unexplored // computes $f(v)$'s magical ordering TopoSort(G^{rev}) // finds SCCs in reverse topological order mark all vertices of G as unexplored $numSCC \leftarrow 0$ foreach $v \in V$ in increasing order of $f(v)$ do if v is unexplored then $numSCC \leftarrow numSCC + 1$ DFS-SCC(G, v)

DFS-SCC
Input: directed graph $G = (V, E)$ in adjacency list representation, and a vertex $s \in V$. Output: every vertex reachable from s is marked as “explored” and has an assigned scc-value. mark s as explored $scc(s) \leftarrow numSCC$ foreach edge (s, v) in s's outgoing adjacency list do if v is unexplored then DFS-SCC(G, v)

Runtime: $O(m + n)$ where $m = |E| \wedge n = |V|$

Obvious Optimization - run DFS on the original graph, just go backwards on the edges.